Reset View

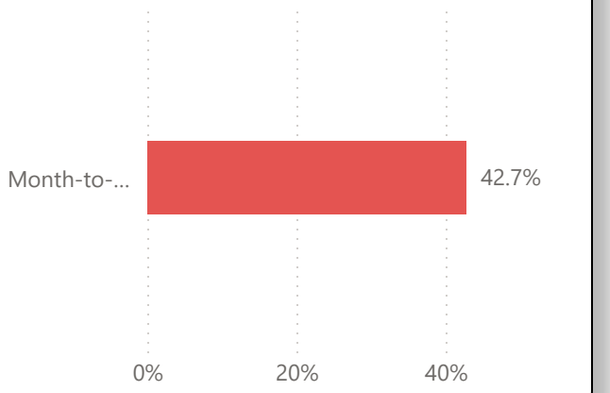
High Risk Segment

Retention Overview — Month-to-month

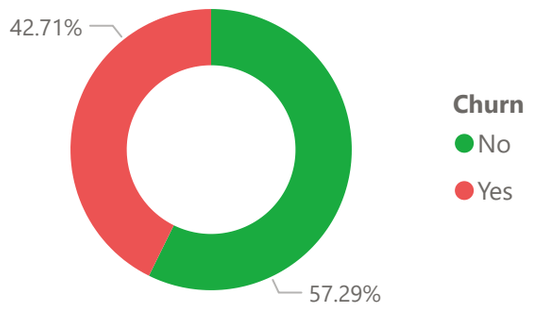
Contract
Month-to-month

Total Customers	Churned Customers	Churn Rate	MRR at Risk	Annual Revenue at Risk
3,875	1,655	42.7%	$120.85K	$1.45M

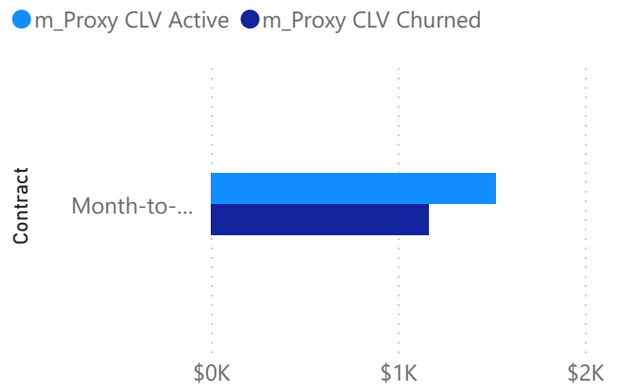
Churn Rate by Contract Type



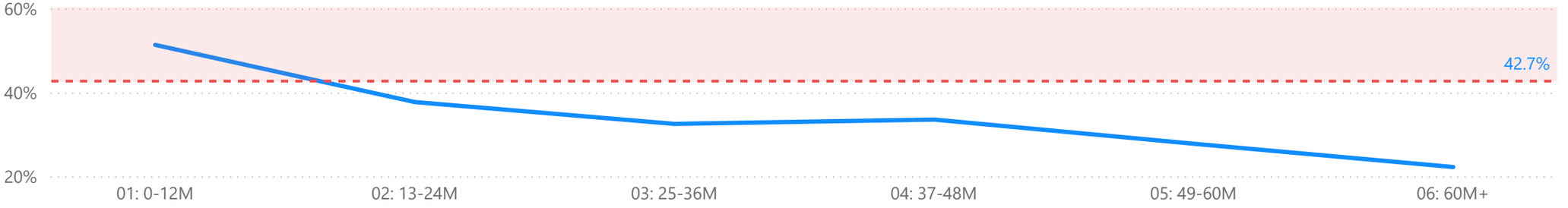
Customer Status



Proxy CLV: Active vs Churned



Churn Rate by Tenure Band



Avg Tenure (Churned)
14.0

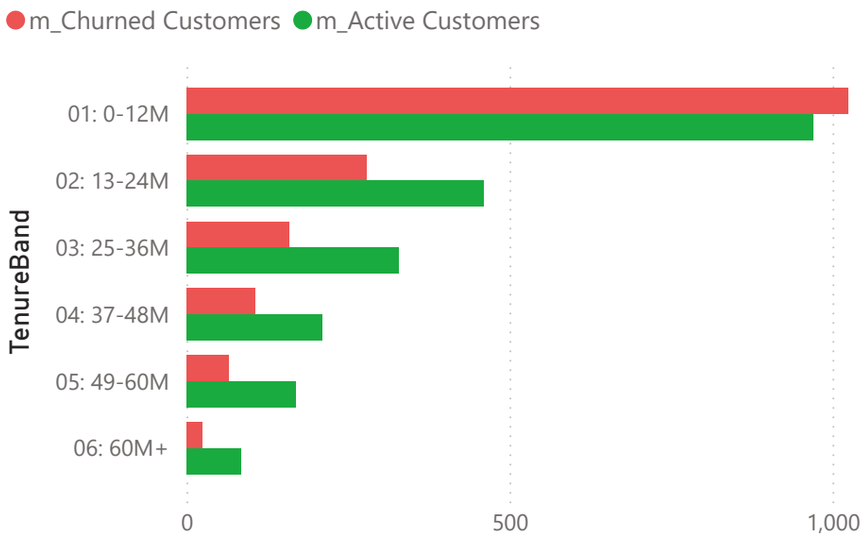
Avg Tenure (Active)
21.0

Contract
Month-to-month

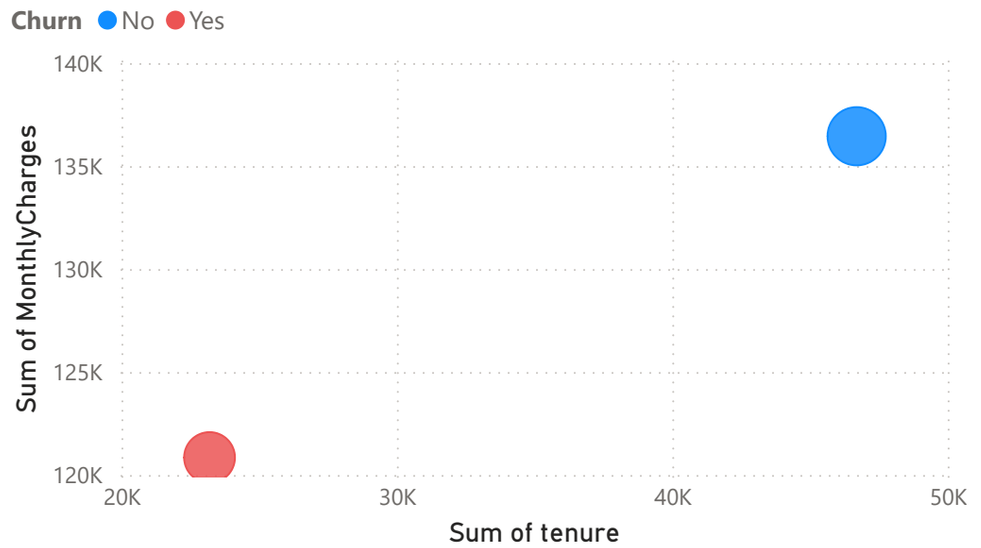
Churn Rate by Tenure & Contract

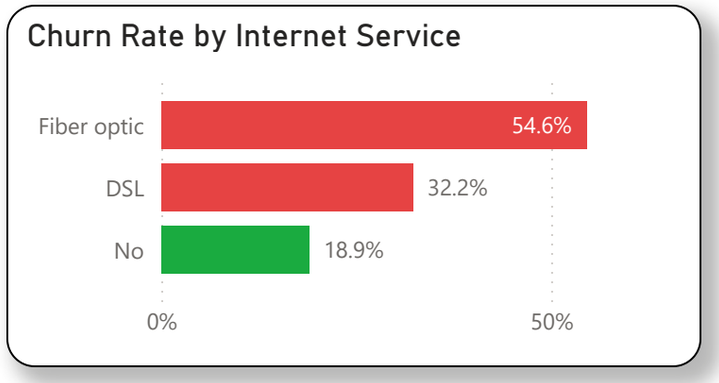
TenureBand	Month-to-month
01: 0-12M	51.4%
02: 13-24M	37.7%
03: 25-36M	32.5%
04: 37-48M	33.5%
05: 49-60M	27.8%
06: 60M+	22.2%

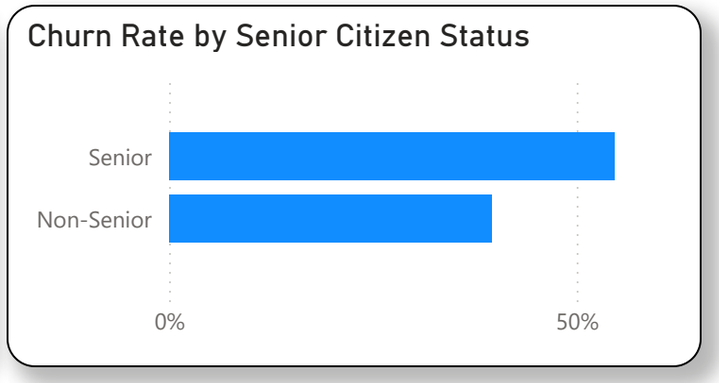
Customer Distribution by Tenure

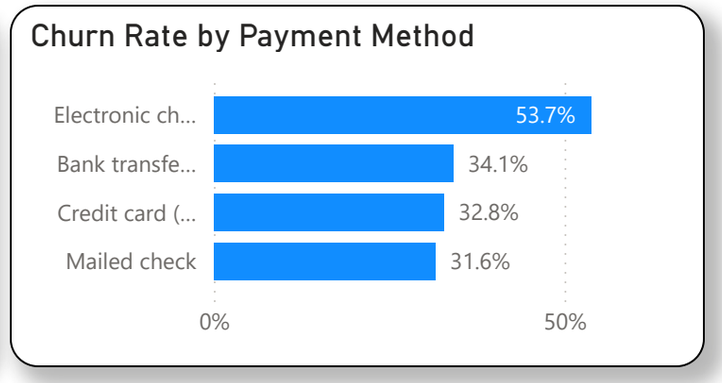


Tenure vs Monthly Charges by Churn Status









OnlineSecurity
- [ ] Select all
- [ ] No
- [ ] Yes

StreamingTV
- [ ] Select all
- [ ] No
- [ ] Yes

TechSupport
- [ ] Select all
- [ ] No
- [ ] Yes

Annual Impact of 5% Churn Reduction

$65.93K

Monthly Charges	Lifetime Value (Proxy)	Tenure (Months)	Churn Status
$185.18K	$1.97K	46K	No

customerID	Contract	InternetService	PaymentMethod	Sum of tenure	Sum of MonthlyCharges	Sum of TotalCharges	Churn	c_TenureBand
0004-TLHLJ	Month-to-month	Fiber optic	Electronic check	4	73.90	280.85	Yes	01: 0-12M
0011-IGKFF	Month-to-month	Fiber optic	Electronic check	13	98.00	1,237.85	Yes	02: 13-24M
0013-EXCHZ	Month-to-month	Fiber optic	Mailed check	3	83.90	267.40	Yes	01: 0-12M
0018-NYROU	Month-to-month	Fiber optic	Electronic check	5	68.95	351.50	No	01: 0-12M
0021-IKXGC	Month-to-month	Fiber optic	Electronic check	1	72.10	72.10	No	01: 0-12M
0023-XUOPT	Month-to-month	Fiber optic	Electronic check	13	94.10	1,215.60	Yes	02: 13-24M
0027-KWYKW	Month-to-month	Fiber optic	Electronic check	23	83.75	1,849.95	No	02: 13-24M
0058-EVZWM	Month-to-month	Fiber optic	Bank transfer (automatic)	55	89.80	4,959.60	No	05: 49-60M
0060-FUALY	Month-to-month	Fiber optic	Electronic check	59	94.75	5,597.65	No	05: 49-60M
0064-YIJGF	Month-to-month	Fiber optic	Bank transfer (automatic)	27	75.75	1,929.00	No	03: 25-36M
0082-OQIQY	Month-to-month	Fiber optic	Electronic check	29	94.20	2,607.60	No	03: 25-36M
0089-IIQKO	Month-to-month	Fiber optic	Credit card (automatic)	39	99.95	3,767.40	No	04: 37-48M
0093-EXYQL	Month-to-month	Fiber optic	Electronic check	40	91.55	3,673.60	No	04: 37-48M
0093-XWZFY	Month-to-month	Fiber optic	Credit card (automatic)	40	104.50	4,036.85	Yes	04: 37-48M
0094-OIFMO	Month-to-month	Fiber optic	Electronic check	11	95.00	1,120.30	Yes	01: 0-12M
0107-YHINA	Month-to-month	Fiber optic	Electronic check	1	99.75	99.75	Yes	01: 0-12M
0111-KLBQG	Month-to-month	Fiber optic	Mailed check	32	93.95	2,861.45	No	03: 25-36M
0112-QAWRZ	Month-to-month	Fiber optic	Bank transfer (automatic)	16	90.80	1,442.20	No	02: 13-24M
0114-PEGZZ	Month-to-month	Fiber optic	Electronic check	33	107.55	3,645.50	No	03: 25-36M
0115-TEEPT	Month-to-month	Fiber optic	Electronic check	21	111.20	2,217.10	Yes	02: 13-24M
Total				**45869**	**185,181.10**	**4,191,552.55**		

Definitions & About

KPI Definitions

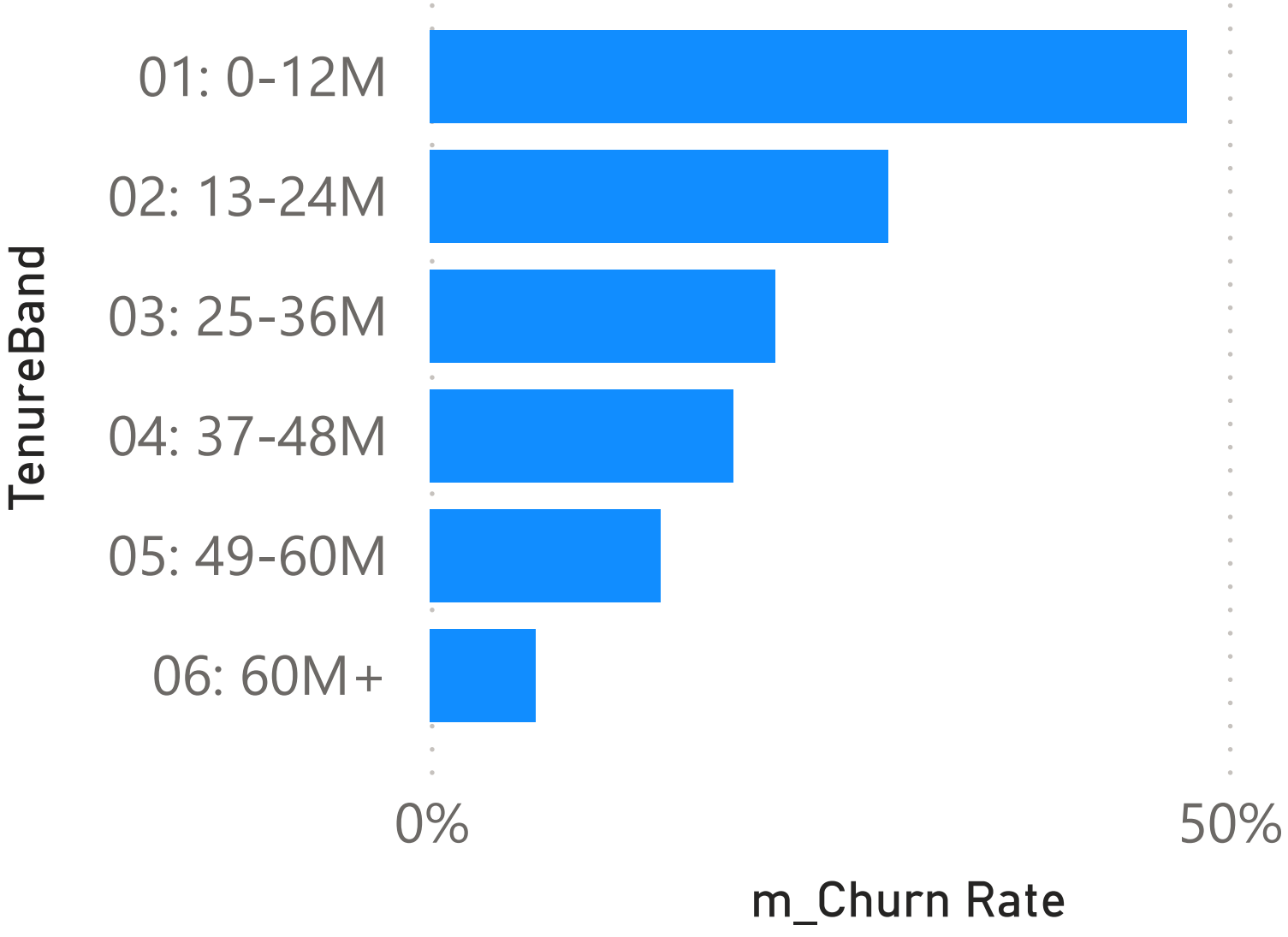
KPI	Definition	Measure Name	Denominator Note
Total Customers	Count of all customers in the dataset regardless of churn status	m_Total Customers	No filter applied
Proxy CLV Churned	Proxy CLV for churned customers only	m_Proxy CLV Churned	Shows lifetime value realised before churn
Proxy CLV Active	Proxy CLV for active customers only	m_Proxy CLV Active	Shows projected lifetime value if retained
Proxy CLV	Average of tenure × monthly charges per customer	m_Proxy CLV	Proxy only — no discount rate or renewal probability applied
MRR at Risk	Sum of monthly charges for churned customers only	m_MRR at Risk	Represents monthly revenue already lost
Cohort Churn Rate	Churn rate calculated within tenure band context only	m_Cohort Churn Rate	Removes all filters except TenureBand
Churned Customers	Count of customers where Churn = Yes	m_Churned Customers	Filtered by c_IsChurned = 1
Churn Rate	Churned customers as a percentage of total customers	m_Churn Rate	Denominator = all customers (snapshot rate, not period rate)
ARPU	Average monthly revenue per customer across all customers	m_ARPU	Denominator = total customers including churned
Annual Revenue at Risk	MRR at Risk multiplied by 12	m_Annual Revenue at Risk	Annualised estimate — assumes no replacement customers
Active Customers	Count of customers where Churn = No	m_Active Customers	Filtered by c_IsChurned = 0
5% Churn Reduction Impact	Estimated annual revenue saved by reducing churn by 5%	m_5pct Churn Reduction Impact	Based on current ARPU × saved customers × 12 months

DATA SOURCE
Dataset: IBM Telco Customer Churn
Source: Kaggle (blastchar)
URL: kaggle.com/datasets/blastchar/telco-customer-churn
Licence: CC0 — Public Domain
Rows: 7,043 customers | Columns: 21

DATA MODEL
Schema: 1 fact table + 3 dimension tables
Fact table: factSubscription (customer snapshot grain)
Dimensions: dimCustomer, dimTenureBand, dimContract
Note: No date table - dataset is a point-in-time snapshot
dimCustomer → factSubscription: bidirectional (1:1 grain)

CHURN RATE DENOMINATOR NOTE
Churn Rate = Churned Customers / Total Customers
Denominator includes ALL customers (active + churned)
This is a snapshot churn rate, not a period churn rate

KNOWN LIMITATIONS
1. Dataset is a point-in-time snapshot - no true time series
2. Cohort analysis uses tenure bands as proxy for time cohorts
3. CLV is a proxy only (tenure × monthly charges) - no discount rate or renewal probability model applied
4. No usage data available - behavioural churn signals not possible
5. Targets table not applicable for this project

ABOUT THIS REPORT
Built by: Sameer Shaik
Role target: Mid-Level Data Analyst
Tools: Power BI Desktop, DAX, Power Query M, GitHub
Dataset: IBM Telco Customer Churn (CC0)
GitHub: github.com/iamsameershaik/pbi-portfolio
LinkedIn: linkedin.com/in/iamsameershaik

Churn by Tenure